System Validation Form

Multi-Game Social Platform with WebSocket Relay Multiplayer

Project Information

Project Title: Godot 4.4 Multi-Game Social Platform with Firebase Authentication and WebSocket Relay Multiplayer

Team Members:

- Name: ______________________ Student ID: ___________
- Name: ______________________ Student ID: ___________
- Name: ______________________ Student ID: ___________

Validator Name: ___________________________________

Validator Position/Title: __________________________

Validation Date: //____

Institution/Company: ______________________________

PART A: PURPOSE OF VALIDATION

1. Validation Objectives

This validation aims to:

- ☑ **Verify Technical Implementation**: Confirm that the system meets all technical specifications and functional requirements
- ☑ **Assess User Experience**: Evaluate the usability, performance, and reliability of the platform
- ☑ **Validate Architecture**: Ensure the WebSocket relay multiplayer architecture works across different network configurations
- ☑ **Confirm Security**: Verify Firebase authentication and data security measures
- ☑ **Test Multiplayer Functionality**: Validate real-time multiplayer gameplay without port forwarding
- ☑ **Evaluate Scalability**: Assess system performance under various network conditions

2. Scope of System Being Validated

Core Components:

1. **Authentication System** - Firebase Auth with Google OAuth integration
2. **Social Hub (Landing)** - User profiles, chat, friend list, online presence
3. **Game Lobbies** - Room creation, browsing, and matchmaking
4. **WebSocket Relay Server** - Production deployment on Render.com
5. **Game #1: Code Breaker** - 1v1 typing battle arena with real-time sync

6. **Game #2: Akashic TCG** - Turn-based card game (in development)
7. **Tutorial System** - Interactive cybersecurity education modules

Technical Stack:

- Frontend: Godot Engine 4.4
- Backend: Node.js Express + express-ws
- Database: Firebase Realtime Database + Firestore
- Authentication: Firebase Auth
- Multiplayer: WebSocket relay architecture
- Deployment: Render.com (production server)

PART B: EVALUATION CRITERIA & RUBRIC

Scoring Scale

- **5** - Exceeds Expectations (Outstanding implementation)
- **4** - Meets Expectations (Fully functional, minor improvements possible)
- **3** - Partially Meets (Functional with some issues)
- **2** - Below Expectations (Significant issues present)
- **1** - Does Not Meet (Non-functional or major failures)
- **N/A** - Not Applicable / Unable to Test

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1. Authentication & User Management (Weight: 15%)

Criteria	Score (1-5)	Comments
Firebase authentication works correctly	_____	
Google OAuth login functional	_____	
User registration with email/password	_____	
Online/offline presence tracking	_____	
User profile display (avatar, level, stats)	_____	
Session persistence across app restarts	_____	

Subsection Score: _____ / 5

Comments:

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2. Social Features & Chat System (Weight: 10%)

Criteria	Score (1-5)	Comments
Real-time chat functionality	____	
Message delivery and synchronization	____	
Unread message indicators	____	
Friend list management	____	
User search functionality	____	
Chat history persistence	____	

Subsection Score: ____ / 5

Comments:

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3. WebSocket Relay Architecture (Weight: 25%)

Criteria	Score (1-5)	Comments
Cross-network connectivity (no port forwarding)	____	
Relay server stability and uptime	____	
Connection handling (2 players max per room)	____	
Message relay accuracy and timing	____	
Automatic reconnection on disconnect	____	
Works on different networks (WiFi/mobile)	____	
Room lifecycle management (create/join/leave)	____	
Host promotion on host disconnect	____	

Subsection Score: ____ / 5

Comments:

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4. Code Breaker Game - Lobby & Room System (Weight: 15%)

Criteria	Score (1-5)	Comments
Room creation and listing	_____	
Room browsing and refresh (5s polling)	_____	
Join room functionality	_____	
Player ready system	_____	
Room state synchronization	_____	
Heartbeat system (30s keep-alive)	_____	
Leave/delete room logic (3 scenarios)	_____	

Subsection Score: _____ / 5

Comments:

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5. Code Breaker Game - Loading Screen (Weight: 10%)

Criteria	Score (1-5)	Comments
Player synchronization display	_____	
Loading progress indicators	_____	
Ready status communication	_____	
Timeout handling (30s max)	_____	
Countdown before arena start	_____	
Relay client preservation	_____	

Subsection Score: _____ / 5

Comments:

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6. Code Breaker Game - Arena Gameplay (Weight: 20%)

Criteria	Score (1-5)	Comments
3-2-1 countdown system with animations	_____	
Real-time typing mechanics	_____	
Damage system and health tracking	_____	
Score synchronization (0.5s intervals)	_____	
Visual effects (shake, particles, shadows)	_____	
Battle music and audio feedback	_____	
Win/loss detection and handling	_____	
Return to room after game ends	_____	
Network latency handling	_____	

Subsection Score: _____ / 5

Comments:

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7. User Interface & User Experience (Weight: 10%)

Criteria	Score (1-5)	Comments
Visual design consistency	_____	
Navigation clarity and intuitiveness	_____	
Responsive UI elements	_____	
Error messages and feedback	_____	
Loading states and transitions	_____	

Criteria	Score (1-5)	Comments
Accessibility considerations	_____	

Subsection Score: _____ / 5

Comments:

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8. Performance & Stability (Weight: 10%)

Criteria	Score (1-5)	Comments
Application startup time	_____	
Scene transition smoothness	_____	
Memory usage and leak prevention	_____	
Frame rate stability (target: 60 FPS)	_____	
Network latency handling	_____	
Crash resistance and error recovery	_____	

Subsection Score: _____ / 5

Comments:

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9. Data Security & Privacy (Weight: 5%)

Criteria	Score (1-5)	Comments
Secure authentication token handling	_____	
Data encryption in transit	_____	
User data privacy protection	_____	
Secure API endpoints	_____	

Subsection Score: _____ / 5

Comments:

PART C: VALIDATION QUESTIONS

Please provide detailed answers to the following questions:

Technical Architecture

1. Did the WebSocket relay architecture successfully eliminate the need for port forwarding?

☐ Yes ☐ No ☐ Partially

Explanation:

2. How did the system perform across different network configurations (home WiFi, mobile data, public networks)?

3. Were there any connectivity issues or scenarios where the relay failed?

Gameplay Experience

4. Is the Code Breaker gameplay engaging and responsive enough for competitive play?

☐ Yes ☐ No ☐ Needs Improvement

Explanation:

5. Did you experience any synchronization issues between players during gameplay?

User Experience

6. How intuitive is the overall user flow from login → lobby → room → arena?

☐ Very Intuitive ☐ Intuitive ☐ Somewhat Confusing ☐ Confusing

Explanation:

7. Were error messages and feedback clear and helpful?

System Reliability

8. Did you encounter any crashes, freezes, or critical errors?

☐ Yes ☐ No

If yes, please describe:

9. How stable was the relay server connection during extended gameplay?

Recommendations

10. What are the top 3 improvements you would recommend for this system?

1.

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2. ```
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3.

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## PART D: OVERALL ASSESSMENT

### Weighted Score Calculation

| Category | Weight | Score (/5) | Weighted Score |
|----------|--------|------------|----------------|
| Authentication & User Management | 15% | _____ | _____ |
| Social Features & Chat | 10% | _____ | _____ |
| WebSocket Relay Architecture | 25% | _____ | _____ |
| Lobby & Room System | 15% | _____ | _____ |
| Loading Screen | 10% | _____ | _____ |
| Arena Gameplay | 20% | _____ | _____ |
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| UI/UX | 10% | _____ | _____ |
| Performance & Stability | 10% | _____ | _____ |
| Security & Privacy | 5% | _____ | _____ |
| **TOTAL WEIGHTED SCORE** | **100%** | | **_____** |

### Final Grade Conversion

- **4.5 - 5.0** = Excellent (A)
- **4.0 - 4.4** = Very Good (B+)
- **3.5 - 3.9** = Good (B)
- **3.0 - 3.4** = Satisfactory (C+)
- **2.5 - 2.9** = Fair (C)
- **Below 2.5** = Needs Major Revision

**Final Grade:** ___________

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## PART E: VALIDATOR RECOMMENDATION

### Does this system meet the requirements for capstone project completion?

☐ **Approved** - System meets all requirements for capstone completion

☐ **Approved with Minor Revisions** - System is acceptable with minor
improvements

☐ **Conditional Approval** - System requires specific revisions before final
approval

☐ **Not Approved** - System requires major revisions

### Specific Revisions Required (if applicable):
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### Additional Comments & Observations:
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## PART F: SIGNATURES

### Validator Declaration

I hereby certify that I have thoroughly tested and evaluated this system according
to the criteria outlined in this validation form. The scores and comments provided
reflect my professional assessment of the system's functionality, performance, and
overall quality.
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**Validator Signature:** __________________________ **Date:** _____/_____/_____

**Validator Printed Name:** _______________________

**Contact Information:** _______________________

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### Student Acknowledgment

We acknowledge that we have reviewed the validation results and understand the
feedback provided.

**Student 1 Signature:** __________________________ **Date:** _____/_____/_____

**Student 2 Signature:** __________________________ **Date:** _____/_____/_____

**Student 3 Signature:** __________________________ **Date:** _____/_____/_____

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### Academic Advisor/Supervisor Approval

I have reviewed the validation results and approve this assessment for capstone
compliance.

**Advisor Signature:** __________________________ **Date:** _____/_____/_____

**Advisor Printed Name:** _______________________

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**Document Version:** 1.0
**Last Updated:** December 1, 2025
**Project Repository:** github.com/muckfaru/capstone2
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